Ting Li (she/her/hers)

Mailing Address
Department of Astronomy and Astrophysics
University of Toronto
50 St George Street
Toronto, ON M5S 3H4
Canada

Email: ting.li@astro.utoronto.ca
Homepage: http://sazabi4.github.io
https://s5collab.github.io/

Research Interests

Near-field Cosmology, Galactic Archaeology, Dark Matter, Dwarf Galaxies, Stellar Streams, Metal-poor Stars, Stellar Populations, Ground-based Instrumentation, Multi-object Spectrograph, Big Data and Survey Science.

Employment

2021 –	Assistant Professor Department of Astronomy and Astrophysics, University of Toronto Toronto, Canada
2019 – 2021	NASA Hubble Fellowship Program Einstein Fellow, Carnegie-Princeton Fellow Carnegie Observatories, Carnegie Institution for Science Pasadena, California, USA Department of Astrophysical Sciences, Princeton University Princeton, New Jersey, USA
2016 – 2019	Leon M. Lederman Fellow in Experimental Physics Fermi National Accelerator Laboratory Batavia, Illinois, USA
2011 – 2016	Research Associate Department of Physics & Astronomy, Texas A&M University College Station, Texas, USA

Education

2010 - 2016	**Ph.D., Physics**, Texas A&M University, Texas, USA Thesis: *Exploring Milky Way Halo Substructures with Large-area Sky Surveys* Advisors: *Dr. Darren L. DePoy, Dr. Jennifer L. Marshall* (Co-chair)
2008 - 2010	**M.S., Space Science and Technology** (SpaceMaster – Erasmus Mundus Course) Luleå University of Technology, Kiruna, Sweden Université Paul Sabatier Toulouse III, Toulouse, France Thesis: *Design of the High Energy Particle Instrument for Electrons for the Energization and Radiation in Geospace Mission* Advisor: *Masafumi Hirahana, University of Tokyo / JAXA*
2004 - 2008	**B.S., Physics,** Fudan University, Shanghai, China Thesis: *Study of Quasar Accretion Disk with Microlensing* Advisor: *Feng Yuan, Shanghai Astronomical Observatory*
2004 - 2008	**Minor, Diplomacy**, Fudan University, Shanghai, China

Grant

Science Projects

2026 – 2028 RCSA Scialog Award Early Science with the LSST (co-PI, **$60,000 USD**)
Measuring Metallicities of RRLs in the Milky Way Vicinity

2026 – 2029 NSERC Alliance International Collaboration Grant (PI, **$282,000 CAD**)
Probing Dark Matter with Astrophysical Observations

2025 – 2027 Canadian Space Agency JWST CYCLE 3 AO 2024 (PI, **$100,000 CAD**)
A Closer Look at the Formation and Evolution of M31's Inner Disk

2023 – 2024 Canada NSERC Alliance International Catalyst Grant (PI, **$25,000 CAD**)
Towards a Unified Framework for Dwarf Galaxy Kinematics

2023 – 2025 Univ. of Toronto Data Science Institute Catalyst Grant (Co-PI, **$100,000 CAD**)
Spectroscopy by the Millions: A Fast, Reproducible Framework to Yield Chemical Compositions of 4 Million Stars

2022 – 2024 Univ. of Toronto Connaught New Researcher Grant (PI, **$20,000 CAD**)
Probe the Nature of Dark Matter with Milky Way's Satellite Galaxies

2022 – 2026 Australia research Council (Co-I, **$700,000 AUD**)
Seeing Dark with Light: Revealing the Milky Way with Stellar Stream

2022 – 2027 Canada NSERC Discovery Grant (PI, **$155,000 CAD**)
Near Field Cosmology w/ Milky Way's Satellite Galaxies & Stellar Streams

2019 – 2021 NASA Hubble Fellowship Program (Science PI, **$340,000 USD**)
Constraining Dark Matter with Stellar Streams and Dwarf Galaxies

Instrumentation Projects

2026 – Canada CFI/ORF Grant (Toronto & Ontario PI, **$2.5M + $2.5M CAD to UToronto**)
A Canadian Contribution to the ANDES Instrument for the ELT

2024 – 2026 Dunlap Seed Fund (Admin-PI, **$30,000 CAD**)
DMD-MOS Development for Astronomical Observation
*DMD-MOS: Digital Micromirror Device-based Multi-Object Spectrograph

2023 – 2025 Univ. of Toronto XSeed Joint Seed Grant (Co-PI, **$120,000 CAD**)
Constructing the human olfactory system ex vivo

2023 – 2025 Dunlap Seed Fund (Co-PI, **$75,000 CAD**)
Investigating Scientific CMOS Detectors for Astronomical Application

2022 – 2027 Canada CFI/JELF Grant (PI, **$350,000 CAD**)
An Advanced Optical Instrumentation Laboratory For the Development of the Next Generation Spectroscopic Surveys

Honors & Awards

2025 OCPA Outstanding Young Researcher Award (OYRA Ardentec Prize)
2024 Scialog Fellow: Early Science with the LSST
2023 Dorothy Shoichet Women Faculty in Science Award of Excellence
2022 Rising Stars in Physics, Princeton University
2019 – 2021 NASA Hubble Fellowship
2019 – 2021 Carnegie-Princeton Fellowship
2016 – 2019 KICP Associate Fellow, University of Chicago
2016 – 2019 Leon M. Lederman Fellow in Experiment Physics
2015 – 2016 Mitchell Institute Graduate Fellowship
2013 – 2014 Dr. Chia-Lai Wang Memorial Scholarship
2008 – 2010 Erasmus Mundus Scholarship
2008 Graduated with First - Class Student, Fudan Univeristy
2004 – 2008 People's Scholarship, Fudan University

Mentorship

Only students and postdoctoral researchers with first-author publications completed under my direct supervision are listed here. Many of my former undergraduates have gone on to pursue graduate studies at leading institutions such as MIT, Yale, UC Berkeley, Caltech, and UArizona.

2024 – Nathan Sandford (Postdoc at UToronto)
- Sandford, Li, et al, ApJ, 2026

2023 – Gustavo Medina (Postdoc at UToronto)
- Medina, Li, et al, ApJ in press, 2025a
- Medina, Li, et al, ApJ in press, 2025b
- Medina, Li, et al, ApJ submitted, 2025c

2024 – Nasser Mohammed (Graduate at UToronto)
- Mohammed, Tang, Li, et al, ApJ in press, 2026

2023 – Emma Jarvis (Graduate at UToronto)
- Jarvis, Li, et al, submitted, 2026

2022 – Mairead Heiger (Graduate at UToronto)
- Heiger, Ji, Speagle, Li, et al, ApJ in press, 2026
- Heiger, Ji, Li, ApJ, 2026
- Heiger, Li, ApJ, 2024

2023 – 2025 Petra Awad (former graduate at U of Groningen; postdoc at Leiden since 2025)
- Awad, Li, et al. A&A, 2025

2023 – 2025 Adi Khandelwal (former undergrad at UToronto; graduate at UArizona since 2025)
- Khandelwal, ..., Li, et al. SPIE, 2024

2023 – 2025 Peter Ma (former undergrad at UToronto; graduate at Berkeley since 2024)
- Ma, Rogers, Li, et al. ApJ, 2025

2022 – 2024 Grace Yu (former undergrad at UToronto; graduate at Caltech since 2024)
- Yu, Li, et al. ApJ, 2024

2021 – 2023 Jordan Bruce (former undergrad at UToronto; graduate at Indiana since 2023)
- Bruce, Li, ApJ, 2023

2020 – William Cerny (former undergrad at UChicago; graduate at Yale since 2022)
- Cerny, Li, et al, submitted, 2026
- Cerny, Drlica-Wagner, Li, ApJ, 2023
- Cerny, Simon, Li, ApJ, 2023

2018 – 2021 Sydney Jenkins (former undergrad at UChicago; graduate at MIT since 2022)
- Jenkins, Li, et al. ApJ, 2021

2017 – 2021 Nora Shipp (former graduate at UChicago; faculty at UW since 2024)
- Shipp, Li, et al. ApJ, 2019

Public Outreach

2025	Presenter, Starfest (Canada's largest amateur astronomy conference), NYAA
2024	Presenter, Star Talks, Astronomy & Space Exploration Association, U. of Toronto
2022	Instructor, Vision of Science Summer Intern Program
2018 –	Presenter, Astronomy on Tap
2018 – 2019	Lecturer, Fermilab Lifelong Learning Institute (LLI) Program
2017 – 2019	Lecturer, Fermilab Saturday Morning Physics for High School Students
2016 – 2019	Coordinator, Fermilab Saturday Morning Physics for High School Students
2014 – 2016	E/PO representative, Dark Energy Survey
2014	Coordinator, Science Olympiad
2012	Instructor, "Expanding Your Horizons" Workshop
2011 – 2014	Organizer, Texas A&M Physics and Astronomy Festival
2011 – 2013	Organizer, Texas A&M Star Parties

Major Scientific Collaboration Involvement

Dark Energy Spectroscopic Instrument (DESI)

• Milky Way Working Group (WG) Chair 2022-now – a WG with about 50 active members

• Builder since 2023 – for the contribution to the DESI focus and alignment system, commissioning and science verification.

• Member since 2016

Dark Energy Survey (DES)

• Milky Way Working Group Chair 2018-2021 – a WG with about 40 active members

• Builder since 2015 – for the contribution to the DES calibration systems, commissioning, operation, and data release

• Member since 2012

Southern Stellar Stream Spectroscopic Survey (S5)

• Leader/Founder since 2018 – a group of about 40 active members

Euclid Consortium

• Member since 2023

SDSS-V

• Member since 2021

DECam Local Volume Exploration (DELVE)

• Member since 2018, Builder since 2025

LSST Dark Energy Science Collaboration (DESC)

• Member since 2016

Service and Leadership

2025 –	Chair, CFHT Community Survey Working Group (CSWG)
2023 – 2027	Canadian representative on the CFHT SAC
2023 – 2025	CADC (Canadian Astronomy Data Centre) User Board
2023	Roman 2022 Peer Review Panelist
2022 –	CASTOR Science Team Member
2022 – 2024	Canadian TAC (Time Allocation Committee) for Gemini and CFHT
2022	HST Cycle 30 Peer Review Panelist
2021 – 2022	DESI Publication Board
2021 –	Canadian Rep on the MSE Science Advisory Group
2021 –	TMT/WFOS Science Team Member
2018 – 2022	Group Leader, Dark Matter Group for Maunakea Spectroscopic Explorer
2017 – 2019	Group Leader, DES Chromatic Correction and Interstellar Reddening Task Force
2016 – 2018	DES Early Career Scientist Committee
2015 –	Referee/reviewer for Nature, ApJ, ApJL, MNRAS, A&A, PASP

Observation, Data Reduction & Instrumentation Experience

Contribution to DES:

- Commissioning and Operation, Photometric Calibration, Data Release
- Designed and built the Atmospheric Transmission Monitoring Camera (aTmCam), including 100+ nights of prototype testing and instrument commissioning
- Earned **DES builder status (authorship)** and **personal data rights** since 2015

Contribution to DESI:

- Active Optics System on Focus and Alignment for DESI
- Target selection and survey planning for Milky Way Survey
- Earned **DESI external participant status** since 2019
- Earned **DESI Builder status** since 2023

Assembled over 80 Instrument Collimators of the Visible Integral-Field Replicable Unit Spectrographs for the Hobby-Eberly Telescope Dark Energy Experiment (HETDEX)

Extensive observing experience:
- Optical imaging: Blanco(4m)/DECam (80+ nights)
- Optical spectroscopy: Magellan(6.5m)/IMACS (60+ nights), Magellan(6.5m)/MIKE (20+ nights), Magellan(6.5m)/M2FS (8 nights); McDonald Observatory, 2.1m/ES2 (30+ nights), 2.7m/VIRUS-P (9 nights).
- **Served as PI or co-PI for over 100 nights on 4m–8m class telescopes, including 100+ nights on Anglo-Australian Telescope, 40+ nights on Magellan, and 100+ hours on VLT**

Selected Recent Conferences, Seminars, Colloquia (past 6 years)

59. SOC and Organizer, Workshop "Near-Field Cosmology in the Era of Big Data: Local Group and Beyond", Fields Institute, Toronto, Canada, Jul 2026
58. SOC, Spec-S5 Dark Matter Meeting, Chicago, USA, Nov 2025
57. Invited Talk, IAU Symposium 403 "The Hidden Beauty of the Galactic Outskirts", Cordoba, Spain, Oct 2025
56. Astro Seminar, Division of Astrophysics | Division of Astrophysics, Lund University, Sweden, Jun 2025
55. SOC and contributed talk, Valencia Workshop on "the Small-Scale Structure of the Universe and Self-Interacting Dark Matter", Valencia, Spain, Jun 2025
54. SOC, CFHT User's Meeting, Montreal, Canada, May 2025
53. Invited talk, Workshop "Cosmological Probes of New Physics", Notre Dame, USA, April 2025
52. Colloquium, Department of Physics & Astronomy, McMaster University, Canada, Feb, 2025
51. Invited Review Talk, IAU Symposium 395 "Stellar populations in the Milky Way and beyond", Paraty, Brazil, Nov 2024
50. Colloquium, Department of Astronomy, University of Arizona, Tucson, AR, Nov 2024
49. Invited Talk, Conference "Galphases24: Accretion in the Local Group", Strasbourg, France, Aug 2024
48. Colloquium, Astronomisches Rechen-Institut, Heidelberg University, Germany, Aug 2024
47. Invited Review Talk, KICP Workshop "Dwarf Galaxies, Star Clusters, and Streams in the LSST Era", University of Chicago, IL, Jul 2024
46. Invited long-term participants, KITP Program "Dark Matter Theory, Simulation, and Analysis in the Era of Large Surveys", University of Santa Barbara, CA, Jun 2024
45. SOC, CASCA (Canadian Annual Conference), Toronto, Jun 2024
44. SOC, Workshop "Globular Clusters and Their Tidal Tails: From the Milky Way to the Local Group", Toronto, June 2024
43. Invited Talk, KITP Conference "Cosmic Signals of Dark Matter Physics: New Synergies", University of Santa Barbara, CA, Jun 2024
42. Invited Astroseminar Talk, Waterloo Centre for Astrophysics, University of Waterloo, ON, Mar 2024
41. Colloquium, Department of Astronomy, Ohio State University, Columbus, OH, Jan 2024
40. Colloquium, Department of Astronomy and Physics, Saint Mary's University, Halifax, NS, Nov 2023
39. Colloquium, Department of Physics, Engineering Physics & Astronomy, Queen's University, Kingston, ON, Oct 2023
38. Colloquium, Department of Astronomy, UC Berkeley, Berkeley, CA, Sept 2023
37. Colloquium, IAS/Department of Astrophysical Sciences, Princeton University, Princeton, NJ, Sept 2023
36. Contributed Talk, Workshop "Great Lakes Clusters and Streams", University of Michigan, Ann Arbor, MI, Aug 2023
35. Contributed Talk, Conference "MODEST-23: Star Clusters in the Post-Pandemic Era", Northwestern University, Evanston, IL, Aug 2023
34. Invited Plenary Talk, DESI Collaboration Meeting, Durham University, Durham, UK, Jul 2023
33. Invited Seminar Talk, Durham University, UK, Jul 2023
32. Invited Seminar Talk, University of Surrey, UK, Jul 2023
31. Colloquium, Kapteyn Astronomical Institute, University of Groningen, the Netherlands, Jul 2023
30. Colloquium, Leiden Observatory, Leiden University, the Netherlands, Jul 2023

29. Invited Review Talk, Pollica Workshop "Self-Interacting Dark Matter: Models, Simulations and Signals", Pollica, Italy, Jun 2023
28. Colloquium, National Astronomical Observatories of China (NAOC), Beijing, China, Jun 2023
27. Colloquium, The Kavli Institute for Astronomy and Astrophysics (KIAA), Peking University, Beijing, China, Jun 2023
26. Invited Topical Talk, IAU Symposium 379 "Dynamical Masses of Local Group Galaxies", Potsdam, Germany, Mar, 2023
25. Invited Virtual (Science & Diversity) Seminar, AAS Division Dynamical Astronomy Community Seminar (Link to Talk), Mar 16, 2023
24. Invited Seminar, University of Victoria, Victoria, BC, Feb 2023
23. Invited Seminar, National Research Council of Canada's Herzberg Astronomy and Astrophysics Research Centre, Victoria, BC, Feb 2023
22. Colloquium, Department of Physics, University of Michigan, Ann Arbor, MI, Nov 2022
21. Astrophysics Colloquium, MIT Kavli Institute, Cambridge, MA, Nov 2022
20. Colloquium, Department of Physics, University of Toronto, Toronto, Canada, Sept 2022
19. Contributed Talk, DECam at 10 Years Workshop, Tucson, AZ, USA, Sept, 2022
18. Invited Seminar Talk, Texas A&M University, College Station, TX, Aug 2022
17. Invited Seminar Talk, University of Notre Dame, Notre Dame, IN, Apr 2022
16. (remote) Invited Plenary Talk, TeV Particle Astrophysics 2021 Conference, Chengdu, China, Oct 27, 2021
15. (remote) Colloquium, Indiana University, Bloomington, IN, USA, Oct 19, 2021
14. (remote) Colloquium, National Research Council of Canada's Herzberg Astronomy and Astrophysics Research Centre, Victoria, BC, Sept 28, 2021
13. (remote) Invited Seminar Talk at N3AS (Link to Talk), May 11, 2021
12. (remote) Colloquium, the University of British Columbia, Vancouver, Canada, Apr 19, 2021
11. (remote) Invited Seminar Talk, the University of Kentucky, Lexington, KY, Feb 3, 2021
10. (remote) Invited Seminar Talk, UC Davis, Davis, CA, Jan 28, 2021
9. (remote) Invited Seminar Talk, University College London, London, England, Nov 2 2020
8. (remote) Invited Seminar Talk, McGill Space Institute, Montreal, Canada, Oct 13 2020
7. iPoster, 235th AAS Meeting, Honolulu, Hawaii, Jan 2020
6. Colloquium, UC Santa Cruz, Santa Cruz, CA, Nov 2019
5. Colloquium, University of Toronto, Toronto, Canada, Oct 2019
4. Invited Talk, LSST Dark Matter Workshop, University of Chicago, Chicago, IL, Aug 2019
3. Invited Talk, IAU Symposium 353 "Galactic Dynamics in the Era of Large Surveys", Shanghai, China, Jul 2019
2. Invited Talk, Conference "Science in our own Backyard: Exploring the Galaxy and the Local Group with WFIRST", Caltech, Pasadena, CA, Jun 2019
1. Invited Talk, KITP Workshop "In the Balance: Stasis and Disequilibrium in the Milky Way", Santa Barbara, CA, Apr 2019

Selected Refereed Publications

Summary: 250+ publications, including 10 1st author, 45 2nd/3rd author, 72 with significant contributions (listed below)
h-index: 98
citations: 40000+
For a complete publication list, please refer to:
https://ui.adsabs.harvard.edu/public-libraries/UGFGnC9bTu-vBkgMrqObgg
or
https://scholar.google.com/citations?user=JTGDv7MAAAAJ&hl=en
Papers led by students or postdocs under my (co-)supervision are highlighted with underlines.

First authors

10. **Li, T. S.**, Ji, A. P., Pace, A. B. et al. "S^5: The Orbital and Chemical Properties of One Dozen Stellar Streams", 2022, ApJ, 928, 30

9. **Li, T. S.**, Koposov, S. E., Erkal, D., et al. "Broken into Pieces: ATLAS and Aliqa Uma as One Single Stream", 2021, ApJ, 911, 149

8. **Li, T. S.**, Koposov, S. E., Zucker, D. B., et al. "The Southern Stellar Stream Spectroscopic Survey (S^5): Overview, Target Selection, Data Reduction, Validation, and Early Science", 2019, MNRAS, 490, 3508

7. **Li, T. S.**, Simon, J. D., Kuehn, K., et al., "The First Tidally Disrupted Ultra-Faint Dwarf Galaxy? - Spectroscopic Analysis of the Tucana III Stream", 2018, ApJ, 866, 22.

6. **Li, T. S.**, Simon, J. D., Pace, A. B., et al. "Ships Passing in the Night: Spectroscopic Analysis of Two Ultra-faint Satellites in the Constellation Carina", 2018, ApJ, 857, 145.

5. **Li, T. S.**, Sheffield, A. A., Johnston, K. V., Marshall, J. L., Majewski, S. R. et al., "Exploring Halo Substructure with Giant Stars", 2017, ApJ, 844, 74.

4. **Li, T. S.**, Simon, J. D., Drlica-Wagner, A., Bechtol, K., et al. "Farthest Neighbor: The Distant Milky Way Satellite Eridanus II", 2017, ApJ, 838, 8

3. **Li, T. S.**, DePoy, D. L., Marshall, J. L., Tucker, D. L., Bernstein, G. M., et al. "Assessment of Systematic Chromatic Errors that Impact Sub-1% Photometric Precision in Large-Area Sky Surveys", 2016, AJ, 151, 157

2. **Li, T. S.**, Balbinot, E., Mondrik, N., et al. "Discovery of a Stellar Overdensity in Eridanus-Phoenix in the Dark Energy Survey", 2016, ApJ, 817, 135

1. **Li, T. S.**, Marshall, J. L., Lépine, S., Williams, P., Chavez, J., "Optical BVRI Photometry of Common Proper Motion F/G/K+M Wide Separation Binaries", 2014, AJ, 148, 60

Second or Third authors, mostly led by my mentees

45. Jarvis, E., **Li, T. S.**, Koposov, S. E., et al. "Characterizing the GD-1 Stream with DESI DR2 Data: Thin Stream and Hot Cocoon", 2026, submitted, arXiv:2604.20958

44. Byström, A., Koposov, S. E., **Li, T. S.**, et al. "Constraining the Galactic bar using the M92 stellar stream", 2026, submitted, arXiv:2605.07918

43. Carlberg, R. G., **Li, T. S.**, Jarvis, E., Mohammed, N., et al. "Joint Modeling of GD-1 and C-19 as Old Streams", 2026, submitted, arXiv:2606.15863

42. Mohammed, N., Tang, J. Y., **Li, T. S.**, et al. "The Kinematically Hot, Extremely Metal-Poor C-19 Stellar Stream in DESI DR2", 2026, ApJ, in press

41. Limberg, G., Ji, A. P., **Li, T. S.**, et al. "S^5: Tidal Disruption in Crater 2 and Formation of Diffuse Dwarf Galaxies in the Local Group", 2025, submitted, arXiv:2512.02177

40. Cerny, W., **Li, T. S.**, Pace, A. B., et al. "A Chemodynamical Census of the Milky Way's Ultra-Faint Compact Satellites. I. A First Population-Level Look at the Internal Kinematics and Metallicities of 19 Extremely-Low-Mass Halo Stellar Systems", 2026, submitted, arXiv:2602.17652

39. Heiger, M., Ji, A. P., **Li, T. S.**, et al. "Not-so-heavy metal(s): Chemical Abundances in the Ultra-faint Dwarf Galaxies Eridanus IV and Centaurus I", 2026, ApJ, 1002, 182

38. Medina, G. E., **Li, T. S.**, Eadie, G. M., et al. "The mass of the Milky Way from outer halo stars measured by DESI DR1", submitted, arXiv:2508.19351

37. Medina, G. E., **Li, T. S.**, Allende Prieto, C., et al. "The DESI Y1 RR Lyrae catalog II: The metallicity dependency of pulsational properties...", ApJ, in press, arXiv:2505.10614

36. Medina, G. E., **Li, T. S.**, Koposov, S. E., et al. "The DESI Y1 RR Lyrae catalog I: Empirical modeling of the cyclic variation of spectroscopic properties and a chemodynamical analysis of the outer halo", ApJ, in press, arXiv:2504.02924

35. Yang, Y., Lewis, G. F., **Li, T. S.**, et al. "Epicyclic density variations in the Indus stellar stream", 2026, *MNRAS*, 547, stag488

34. Yang, H., Wang, W., **Li, T. S.**, et al. "The Shape-velocity Alignment of Satellites Forged by Tidal Locking and Dynamical Friction", 2026, *ApJ*, 999, 158

33. Sandford, N. R., **Li, T. S.**, Koposov, S. E., et al. "Chemodynamics of Boötes I with S^5: Revised Velocity Gradient, Dark Matter Density, and Galactic Chemical Evolution Constraints", 2026, *ApJ*, 998, 47

32. Koposov, S., **Li, T. S.**, Allende Prieto, C., et al. "DESI Data Release 1: Stellar Catalogue", 2026, *OJAp*, 9, 55260

31. Ding, J., Rockosi, C., **Li, T. S.**, et al. "The Draco Dwarf Spheroidal Galaxy in the First Year of Dark Energy Spectroscopic Instrument Data", 2025, *ApJ*, 994, 134

30. Pace, A. B., **Li, T. S.**, Ji, A. P., et al. "Spectroscopic Analysis of Pictor II: a very low metallicity ultra-faint dwarf galaxy bound to the Large Magellanic Cloud", 2025, *OJAp*, 8, 112

29. Kim, B., Koposov, S. E., **Li, T. S.**, et al. "Nearby stellar substructures in the Galactic halo from DESI Milky Way Survey Year 1 Data Release", 2025, *MNRAS*, 540, 264

28. Ma, P. X., Rogers, K. K., **Li, T. S.**, et al. "Toward Characterizing Dark Matter Subhalo Perturbations in Stellar Streams with Graph Neural Networks", 2025, *ApJ*, 987, 96

27. Awad, P., **Li, T. S.**, Erkal, D. et al., "S5: New insights from deep spectroscopic observations of the tidal tails of the globular clusters NGC 1261 and NGC 1904", 2025, A&A, 693, A69.

26. Simon, J. D., **Li, T. S.** et al. "Eridanus III and DELVE 1: Carbon-rich Primordial Star Clusters or the Smallest Dwarf Galaxies?", 2024, ApJ, 976, 256S

25. Yu, F., **Li, T. S.**, Speagle, J. S. et al., "The Power of High Precision Broadband Photometry: Tracing the Milky Way Density Profile with Blue Horizontal Branch stars in the Dark Energy Survey", 2024, ApJ, 975, 81Y

24. Hansen, T. T., Simon, J. D., **Li, T. S** et al. "Chemical Diversity on Small Scales: Abundance Analysis of the Tucana V Ultrafaint Dwarf Galaxy", 2024ApJ, 968, 21H.

23. Usman, S. A., Ji, A. P., **Li, T. S.**, et al., “Multiple Populations and a CH Star Found in the 300S Globular Cluster Stellar Stream”, 2024, MNRAS, 529, 2413.

22. Heiger, M. E., **Li, T. S.**, Pace, A. B. et al. "Reading Between the (Spectral) Lines: Magellan/IMACS spectroscopy of the Ultra-faint Dwarf Galaxies Eridanus IV and Centaurus I", 2024, ApJ, 961, 234H.

21. Cerny, W., Drlica-Wagner, A., **Li, T. S.**, et al. "DELVE 6: An Ancient, Ultra-faint Star Cluster on the Outskirts of the Magellanic Clouds", 2023, ApJ, 953L, 21C

20. Koposov, S. E., Erkal, D., **Li, T. S.** et al. "S5: Probing the Milky Way and Magellanic Clouds potentials with the 6D map of the Orphan-Chenab stream", 2023, MNRAS, 521, 4936

19. Bruce, J., **Li, T. S.** Pace, A. B. et al. "Spectroscopic analysis of Milky Way outer halo satellites: Aquarius II and Bootes II", 2023, ApJ, 950, 167B

18. Hansen, T. T., Simon, J. D., **Li, T. S.** et al. "Evidence for multiple nucleosynthetic processes from carbon-enhanced metal-poor stars in the Carina dwarf spheroidal galaxy", 2023, A&A, 674A, 180H

17. Cerny, W., Simon, J. D., **Li, T. S.** et al. "Pegasus IV: Discovery and Spectroscopic Confirmation of an Ultra-Faint Dwarf Galaxy in the Constellation Pegasus", 2023, ApJ, 942, 111

16. Pace, A. B., Erkal, D., **Li, T. S.**, "Proper Motions, Orbits, and Tidal Influences of Milky Way Dwarf Spheroidal Galaxies", 2022, ApJ, 940, 136

15. Ji, A. P., Koposov, S. E., **Li, T. S.**, et al. "Kinematics of Antlia 2 and Crater 2 from the Southern Stellar Stream Spectroscopic Survey (S^5)", 2021, ApJ, 921, 32

14. Jenkins, S., **Li, T. S.**, Pace, A. B., et al. "VLT Spectroscopy of Ultra-Faint Dwarf Galaxies. 1. Boötes I, Leo IV, Leo V", 2021, ApJ, 920, 92J

13. Ji, A. P., **Li, T. S.**, Hansen, T. T., et al. "The Southern Stellar Stream Spectroscopic Survey (S5): Chemical Abundances of Seven Stellar Streams", 2020, AJ, 160, 181

12. Simon, J. D., **Li, T. S.**, Erkal, D., et al. "Birds of a Feather? Magellan/IMACS Spectroscopy of the Ultra-faint Satellites Grus II, Tucana IV, and Tucana V", 2020, ApJ, 892, 137

11. Wan, Z., Lewis, G. F., **Li, T. S.**, et al. "The tidal remnant of an unusually metal-poor globular cluster", 2020, Nature, 583, 768

10. Koposov, S. E., Boubert, D., **Li, T. S.**, et al. "Discovery of a nearby 1700 km/s star ejected from the Milky Way by Sgr A*", 2020, MNRAS, 491, 2465

9. Ji, A. P., **Li, T. S.**, Simon, J. D., et al. "Detailed Abundances in the Ultra-faint Magellanic Satellites Carina II and III", 2020, ApJ, 889, 27J

8. Shipp, N., **Li, T. S.**, Pace, A. B., et al. "Proper Motions of Stellar Streams Discovered in the Dark Energy Survey", 2019, ApJ, 885, 3

7. Koposov, S. E., Belokurov, V., **Li, T. S.**, et al. "Piercing the Milky Way: an all-sky view of the Orphan Stream", 2019, MNRAS, 485, 4726

6. Pace, A. B. & **Li, T. S.**, "Proper motions of Milky Way Ultra-Faint satellites with *Gaia* DR2 $\times$ DES DR1", 2019, ApJ, 875, 77

5. Erkal, D., **Li, T. S.**, Koposov, S. E., Belokurov, V., et al., "Modelling the Tucana III stream - a close passage with the LMC", 2018, MNRAS, 481, 3148.

4. Nagasawa, D. Q., Marshall, J. L., **Li, T. S.** et al. "Chemical Abundance Analysis of Three α-Poor, Metal-Poor Stars in the Ultra-Faint Dwarf Galaxy Horologium I", 2018, ApJ, 852, 99.

3. Simon, J. D., **Li, T. S.**, Drlica-Wagner, A., Bechtol, K., et al. "Nearest Neighbor: The Low-mass Milky Way Satellite Tucana III", 2017, ApJ, 838, 11

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43. Luo, Y., Leauthaud, A., Greene, J., Huang, S., Kado-Fong, E., Danieli, S., **Li, T. S.**, et al. "The Merian Survey: Design, Construction, and Characterization of a Filter Set Optimized to Find Dwarf Galaxies and Measure their Dark Matter Halo Properties with Weak Lensing", 2024, MNRAS, 530, 4988
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19. Simon, J. D., Brown, T. M., Drlica-Wagner, **Li, T. S.**, et al. "Eridanus II: A Fossil from Reionization with an Off-center Star Cluster", 2021, ApJ, 908, 18

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15. Nadler, E. O., Wechsler, R. H., Bechtol, K. et al. (including **Li, T. S.**), "Milky Way Satellite Census. II. Galaxy-Halo Connection Constraints Including the Impact of the Large Magellanic Cloud", 2020, ApJ, 893, 48

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12. Marshall, J. L., Hansen, T., Simon, J. D., **Li, T. S.**, et al. "Chemical Abundance Analysis of Tucana III, the Second r-process Enhanced Ultra-faint Dwarf Galaxy", 2019, ApJ, 882, 177
11. Mau, S., Drlica-Wagner, A., Bechtol, K., Pace, A. B., **Li, T. S.**, et al. "A Faint Halo Star Cluster Discovered in the Blanco Imaging of the Southern Sky Survey", 2019, ApJ, 875, 154
10. Torrealba, G., Belokurov, V., Koposov, S. E., **Li, T. S.**, et al. "The hidden giant: discovery of an enormous Galactic dwarf satellite in Gaia DR2", 2019, MNRAS, 488, 2743
9. Erkal, D., Belokurov, V., Laporte, C. F. P., Koposov, S. E., **Li, T. S.**, et al. "The total mass of the Large Magellanic Cloud from its perturbation on the Orphan stream", 2019, MNRAS, 487, 2685
8. Wang, M. Y., de Boer, T., Pieres, A., **Li, T. S.**, et al. "The Morphology and Structure of Stellar Populations in the Fornax Dwarf Spheroidal Galaxy from Dark Energy Survey Data", 2019, ApJ, 881, 118
7. Wang, M. Y., Koposov, S., Drlica-Wagner, A., Pieres, A., **Li, T. S.**, "Rediscovery of the Sixth Star Cluster in the Fornax Dwarf Spheroidal Galaxy", 2019, ApJ, 875, 13
6. Shipp, N., Drlica-Wagner, A., Balbinot, E., Ferguson, P., Erkal, D., **Li, T. S.**, et al., "Stellar Streams Discovered in the Dark Energy Survey", 2018, ApJ, 862, 114.
5. Bernstein, G. M., Abbott, T. M. C., Armstrong, R., Burke, D. L., Diehl, H. T., Gruendl, R. A., Johnson, M. D., **Li, T. S.**, et al., "Photometric Characterization of the Dark Energy Camera", 2018, PASP, 130, 4501.
4. Bergemann, M., Sesar, B., Cohen, J., Serenelli, A. M., Sheffield, A. A., **Li, T. S.** et al., "Two chemically similar stellar overdensities on opposite sides of the plane of the Galactic disk", 2018, Nature, 555, 334.
3. Hansen, T. T., Simon, J. D., Marshall, J. L., **Li, T. S.**, et al. "An r-process Enhanced Star in the Dwarf Galaxy Tucana III", 2017, ApJ, 838, 44
2. Johnston, K. V., Price-Whelan, A. M., Bergemann, M., Laporte, C., **Li, T. S.**, et al. "Disk Heating, Galactoseismology, and the Formation of Stellar Halos", 2017, Galaxies, 5, 44
1. Melendez, J., Placco, V. M., Tucci-Maia, M., Ramirez, I., **Li, T. S.**, "2MASS J18082002-5104378: The Brightest (V=11.9) Ultra Metal-Poor Star", 2016, A&A, 585, 5

Selected Conference Proceedings

Summary: Most of my conference proceedings are related to instrumentation work and have been published in the SPIE proceedings.

7. Khandelwal, A., et al. (including **Li, T. S.**), "Beyond CCDs: Characterization of sCMOS detectors for optical astronomy", 2024, Proc. of SPIE Vol. 13103, 131030R
6. Meisner, A. M., et al. (including **Li, T. S.**), "Performance of Kitt Peak's Mayall 4-meter telescope during DESI commissioning", 2020, Proc. of SPIE Vol. 11447, 1144794
5. Drlica-Wagner, A., Marrufo Villalpando, E., O'Neil, J., Estrada, J., Holland, S., Kurinsky, N., **Li, T. S.**, "Characterization of skipper CCDs for cosmological applications", 2020, Proc. of SPIE Vol. 11454, 114541A
4. **Li, T. S.**, DePoy, D. L., Marshall, J. L., Nagasawa, D. Q., Carona, D. W., Boada, S., "Monitoring the atmospheric throughput at Cerro Tololo Inter-American Observatory with aTmCam", 2014, Proc. of SPIE Vol. 9147, 91476Z, arXiv:14077047
3. DePoy, D. L., Allen, R., **Li, T. S.**, Marshall, J. L., Papovich, C., Prochaska, T., Shectman, S., "An update on the wide field, multi-object, moderate-resolution, spectrograph for the Giant Magellan Telescope", 2014, Proc. of SPIE Vol. 9147, 914720

2. Marshall, J. L., DePoy, D. L., Prochaska, T., Allen, R.D., Williams, P., Rheault, J.-P., **Li, T. S.**, and 22 colleagues, "VIRUS instrument collimator assembly", 2014, Proc. of SPIE Vol. 9147, 91473S

1. **Li, T. S.**, DePoy, D. L., Kessler, R., Burke, D. L., Marshall, J. L., Wise, J., Rheault, J.-P., Carona, D. W., Boada, S., Prochaska, T., Allen, R., "aTmcam: a simple atmospheric transmission monitoring camera for sub 1% photometric precision", 2012, Proc. of SPIE Vol. 8446, 84462L, arXiv:14077047

Selected White Papers, Reports, Books

7. Facilitator for Snowmass CF03 white papers (arXiv:2203.06200, arXiv:2209.08215)

6. Author for 7 Astro2020 Science White Papers (including one leading author).

5. Chapter Leader, The Detailed Science Case for the Maunakea Spectroscopic Explorer, 2019 edition, 2019, arXiv:1904.04907 (entire document), arXiv: 1903.03155 (one chapter)

4. Chapter Author, Probing the Fundamental Nature of Dark Matter with the Large Synoptic Survey Telescope, 2019, arXiv:1902.01055

3. Organizer and Chapter Author, Petabytes to Science, 2019, arXiv:1905.05116

2. Chapter Author, GMT Science Book 2018: https://www.gmto.org/gallery/gmt-resources/#GMT_Science_Book_2018

1. Chapter Author, Maximizing Science in the Era of LSST: A Community-Based Study of Needed US Capabilities, 2016, arXiv:1610.01661

Student and Postdoctoral Supervision

Former undergraduate mentees who led or co-authored publications are listed earlier under "Mentorship." Listed below are all current supervisees.

Postdoctoral Fellows

Biprateep Dey (2025 –) – Schmidt AI Fellow

Kevin McKinnon (2025 –) – Schmidt AI Fellow

Ryan Dungee (2024 –) – Dunlap Instrumentation Fellow

Gustavo Medina (2023 –) – personal postdoc (2023), A&S Fellow (2023 – 2025), Dunlap Fellow (2025 –)

Nathan Sandford (2023 –) – joint personal postdoc with J. Bovy (2023 – 2024), A&S Fellow (2024 –)

Graduate Students

Anika Slizewski (2025 –) – PhD co-advisor (with Gwen Eadie); *Mass estimators with phase-space distribution functions*

Emma Jiayi Xu (2025 –), Physics Department – Master's thesis co-advisor (with Barth Netterfield); *Optimizing filter selections for GigaBIT*

Nasser Mohammed (2024 –) – PhD advisor; *Constraining dark matter with stellar streams in DESI*

Emma Jarvis (2023 –) – AST 1500 project supervisor; *Characterizing the GD-1 stream with DESI DR2*

Mairead Heiger (2022 –) – PhD co-advisor (with Josh Speagle); *Galactic chemical evolution with dwarf galaxies and metal-poor stars*; awarded a Prize Fellowship from the University of Virginia (from Fall 2026)

Graduate Supervisory Committees (PhD, University of Toronto)

Committee Member: Grace Connolly (2026 –), Qing Liu (2021 – 2024)

Final Oral Examination Committee: Steffani Grondin (2025), Nathaniel Starkman (2024), Aarya Patil (2023)

Thesis Qualifying Exam Committee: Samantha Berek (2022), Steffani Grondin (2022)

Current Undergraduate Research Students (2025 – 2026, University of Toronto)

Rosayla Coulthard (ROP + AST 430) – Carbon-enhanced metal-poor stars in Milky Way halo satellites

Alexandros Pratsos (SUDS + AST 430) – Identifying Milky Way streams with machine learning

Nava Wolfish (SURP + AST 425) – Formation history of Andromeda with JWST

Zain Azam (Work Study) – CMOS detector characterization

Daksh Singh (Work Study) – Optical fiber focal-ratio-degradation and throughput characterization system

Liwen Chen (ROP) – Stellar motions with JWST and Gaia

Mingzhi Jiang (ROP) – RR Lyrae in the Orphan-Chenab stream with DECam

Wesley Luo (SURP/USRA) – High-precision stellar motions in the Local Universe

Aishani Chaudhuri (SURP/UTEA) – Binary stars in DESI multi-epoch spectroscopy

Emma Jiayi Xu (SURP) – Prototyping the next-generation spectrograph for space exploration

Juan Carlos Aranda Muñoz (Mitacs) – Forecasting dwarf-galaxy dark-matter profile constraints for DESI

Alejandra Calderón Linares (Mitacs) – Searching for stellar streams with 10 million Milky Way stars

Isabelle Laing (AST 430; with Josh Speagle) – Galactic paleontology: assembly history from surviving stars

Dhruv Pandya (Work Study) – Digital Micromirror Device-based Multi-Object Spectrograph (DMD-MOS)

Diego Martinez Noriega (Waterloo Co-op; with Josh Speagle) – Fast, reproducible chemical compositions from spectroscopic surveys